Vendor Performance Analysis

Optimizing profitability in retail and wholesale requires effective inventory and sales management. This analysis identifies areas for improvement, from underperforming brands to vendor dependency.

Business Problem: Key Optimization Goals

Identify Underperforming Brands

Pinpoint brands needing promotional or pricing adjustments.

Determine Top Vendors

Recognize vendors driving significant sales and gross profit.

Analyze Bulk Purchasing

Assess its impact on unit cost savings.

Improve Inventory Turnover

Reduce holding costs and enhance efficiency.

Investigate Profitability Variance

Compare high vs. low-performing vendors.

Exploratory Data Analysis: Summary Statistics

Negative & Zero Values

- **Gross Profit:** Minimum of -52,002.78, indicating potential losses.
- **Profit Margin:** Minimum of -∞, suggesting instances where revenue is zero or below cost.
- **Total Sales Quantity & Sales Dollars:** Zero sales for some products, indicating slow-moving or obsolete stock.

Outliers Detected

- **Purchase & Actual Prices:** Max values (5,681.81 & 7,499.99) significantly higher than mean (24.39 & 35.64), indicating premium offerings.
- **Freight Cost:** Extreme variation (0.09 to 257,032.07) suggests logistics inefficiencies.
- **Stock Turnover:** Ranges from 0 to 274.5, showing rapid sales for some, long periods unsold for others.

Data Filtering for Reliable Insights

To enhance the reliability of our insights, inconsistent data points were removed:

Gross Profit ≤ 0

Excluding transactions that resulted in losses.

Profit Margin ≤ 0

Focusing analysis on profitable transactions only.

Total Sales Quantity = 0

Eliminating inventory that was never sold.

Correlation Insights: Key Connections

1. **Purchase & Sales Values**

   Strong positive correlation (0.83–0.98): Higher purchasing leads to higher sales.

2. **Gross Profit & Sales**

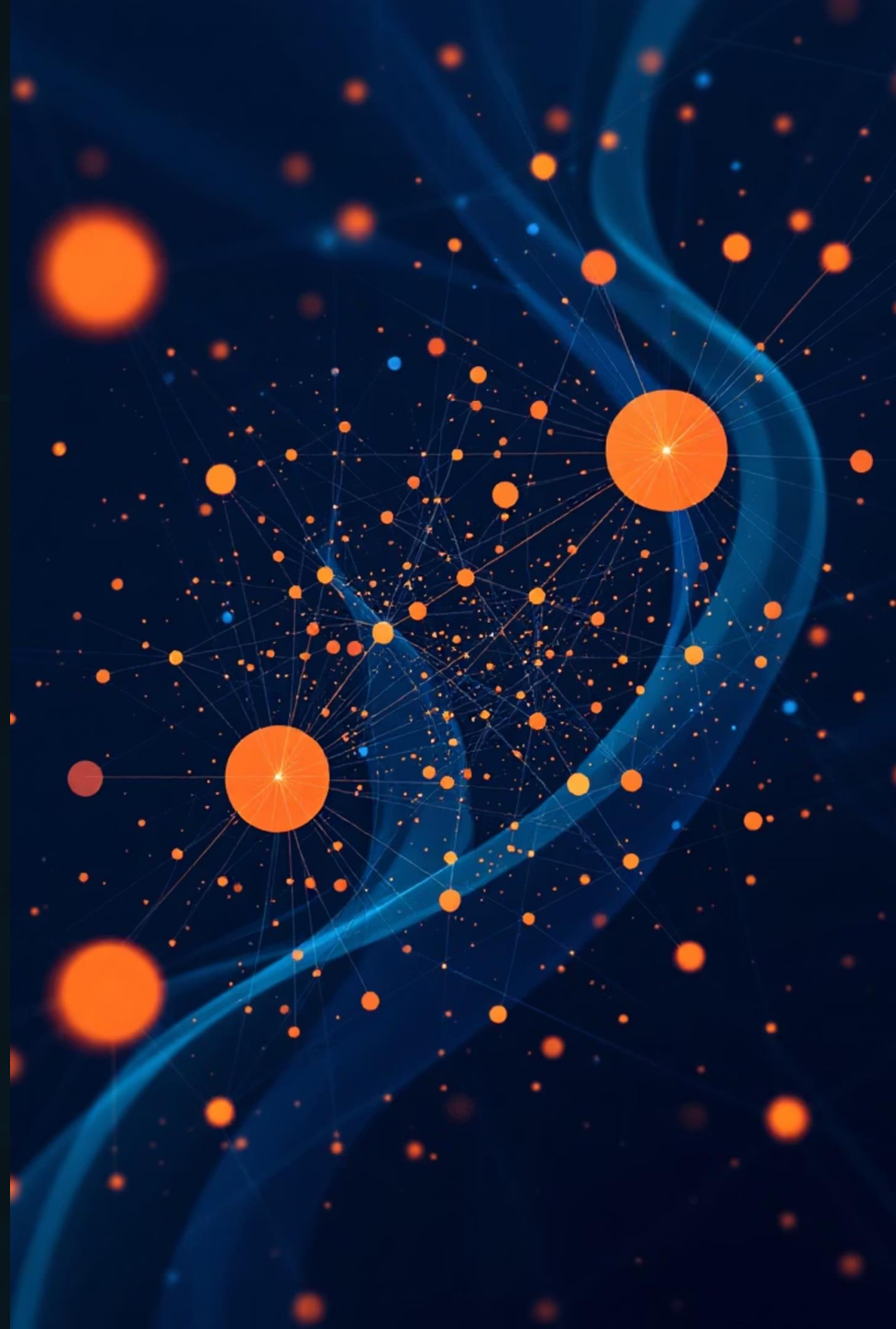
   Strongly correlated (0.98 with Sales Dollars, 0.70 with Sales Quantity): More sales directly increase profit.

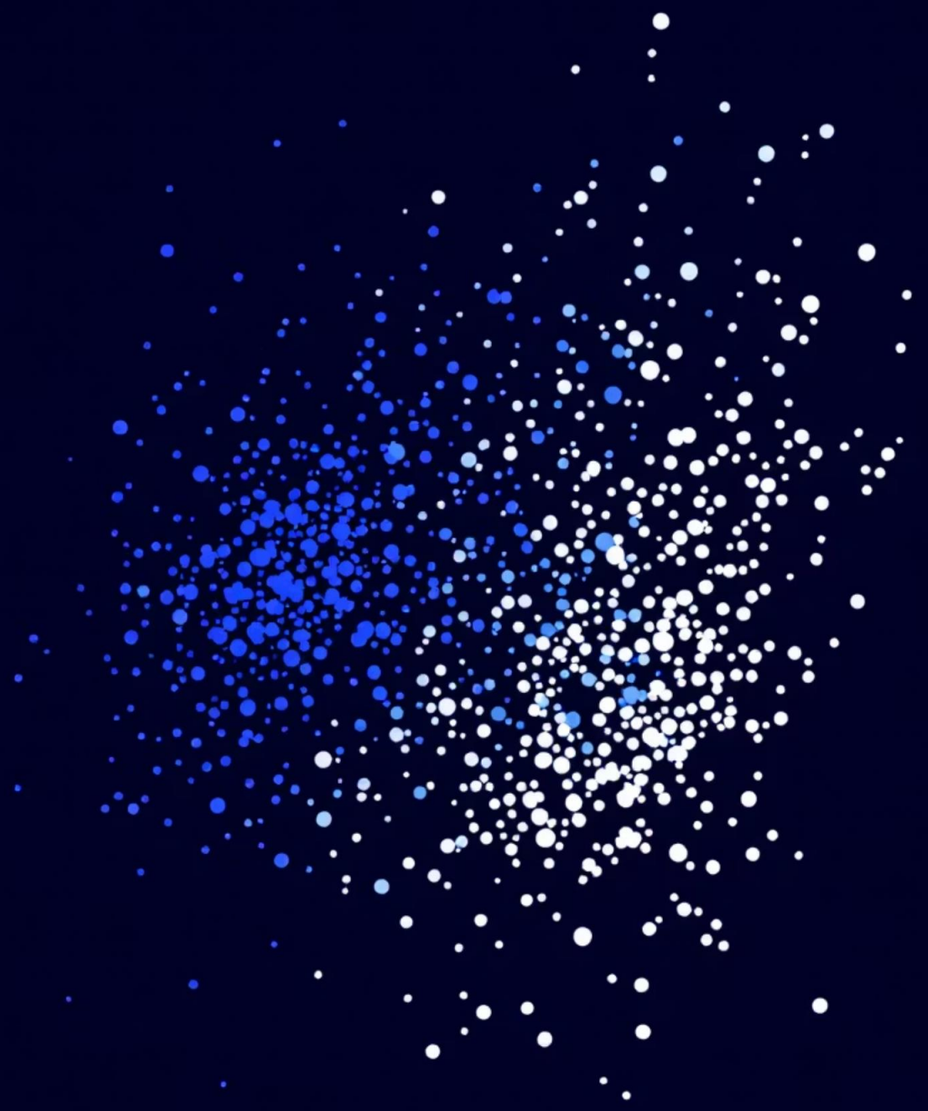
3. **Stock Turnover & Profit Margin**

   Good positive correlation (0.40): Faster-selling products often yield better profit margins.

4. **Freight Cost & Sales/Purchase**

   Positively correlated (0.17-0.20): Shipping costs increase with more stock handled.





Correlation Insights: Further Discoveries

1 Purchase Price & Actual Price

Extremely high correlation (0.99): Minimal price fluctuations, sellers charge almost the same as purchase cost.

2 Brand & Vendor Number

Very low or no correlation with sales/profit: Performance depends more on products and quantities, not brand/vendor ID.

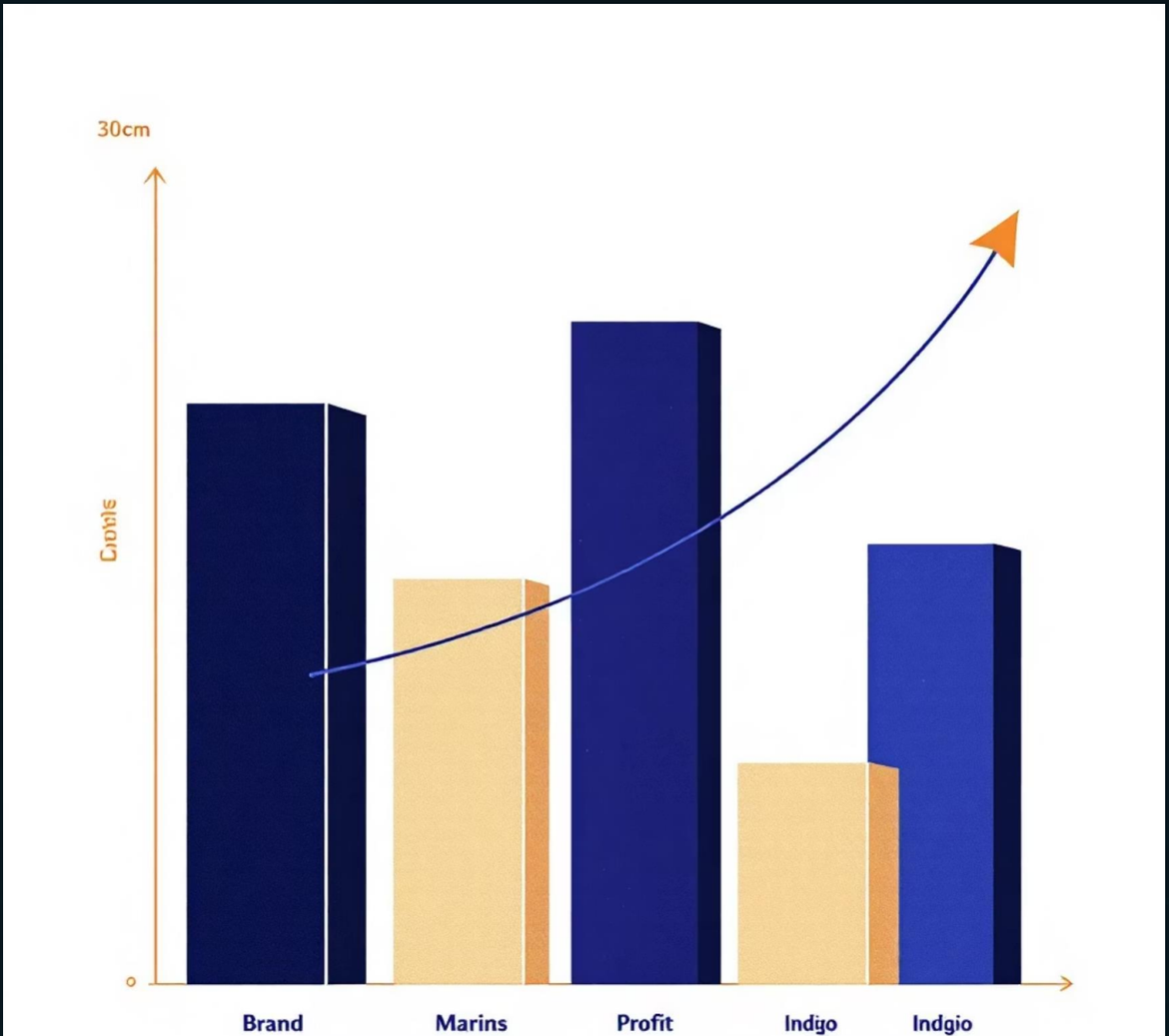
3 Excise Tax & Total Sales

Strong correlation (0.82–0.88): Higher sales lead to higher tax, as expected.

Research Questions & Key Findings

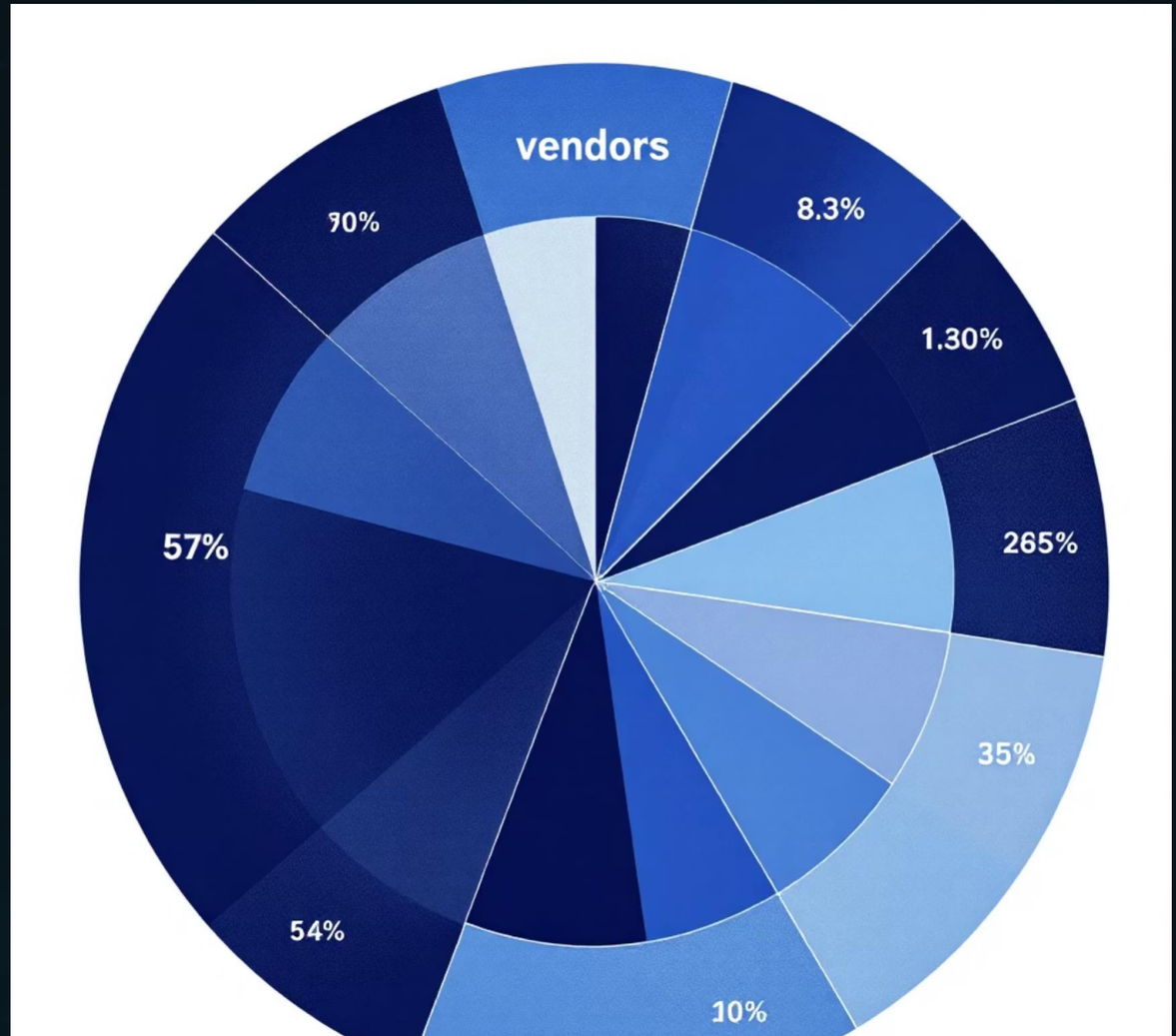
Brands for Promotional Adjustments

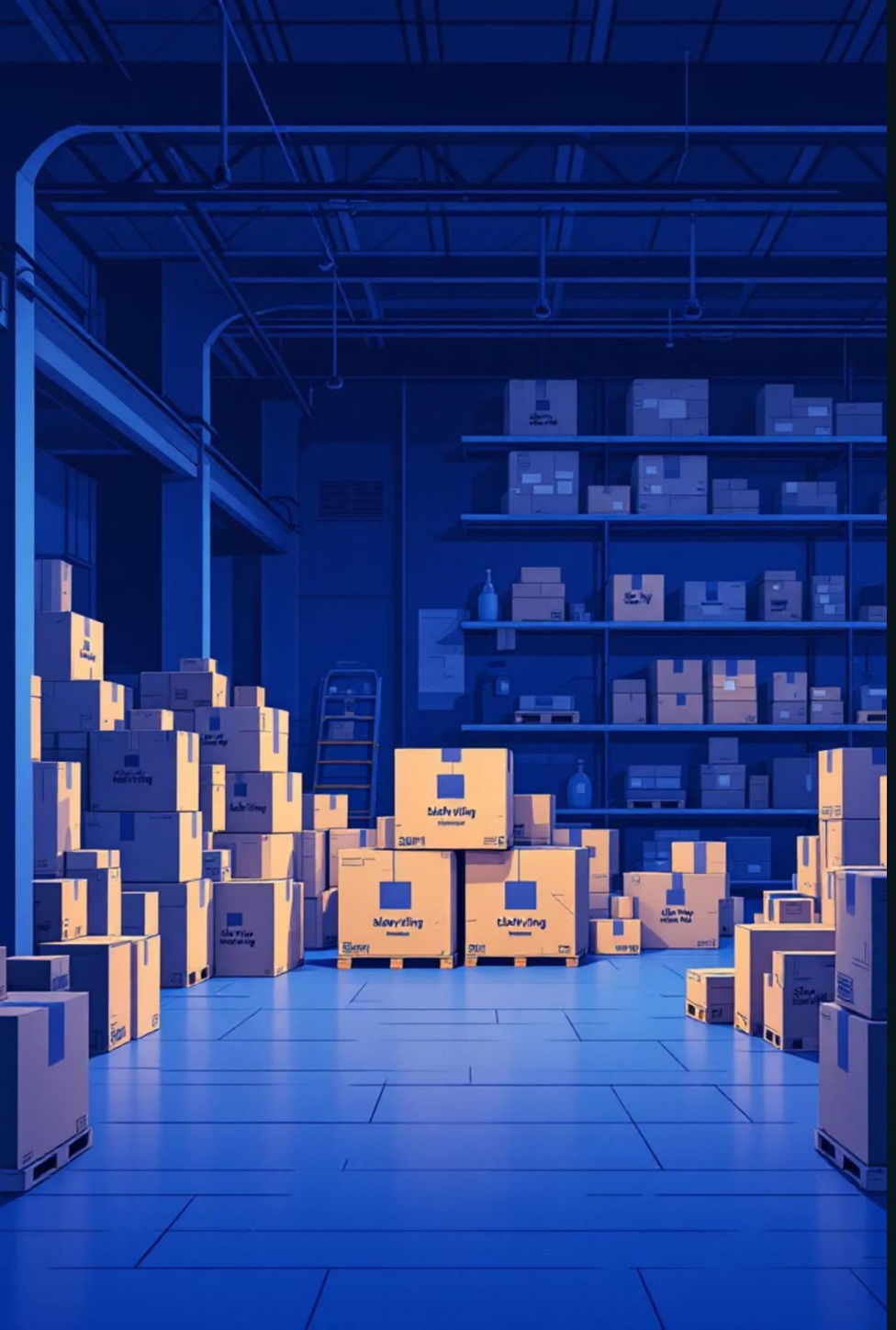
198 brands show lower sales but higher profit margins. These could benefit from targeted marketing or price optimizations to boost volume without compromising profitability.



Top Vendors by Contribution

Top 10 vendors contribute 65.69% of total purchases. This over-reliance introduces supply chain risks, indicating a need for diversification.





Research Questions & Key Findings (Cont.)

1

Bulk Purchasing Impact

Bulk buyers receive 72% lower unit costs ($10.78 vs. higher for smaller orders). This strategy increases sales while maintaining profitability.

2

Low Inventory Turnover

Total Unsold Inventory Capital: $2.71M. Slow-moving inventory increases storage costs and reduces cash flow. Identifying these vendors is crucial for better stock management.

3

Profit Margin Comparison

Top Vendors: Mean 31.17% (CI: 30.74-31.61%). Low Vendors: Mean 41.55% (CI: 40.48-42.62%). Low-performing vendors have higher margins but struggle with sales, suggesting pricing inefficiencies or market reach issues.

Dashboard Insights: Performance Overview

Sales & Purchase Overview

Generated $441.41M in sales with $307.34M purchase cost, yielding $134.07M gross profit. Profit margin at 331.62K indicates efficient performance.

Vendor Contribution & Dependency

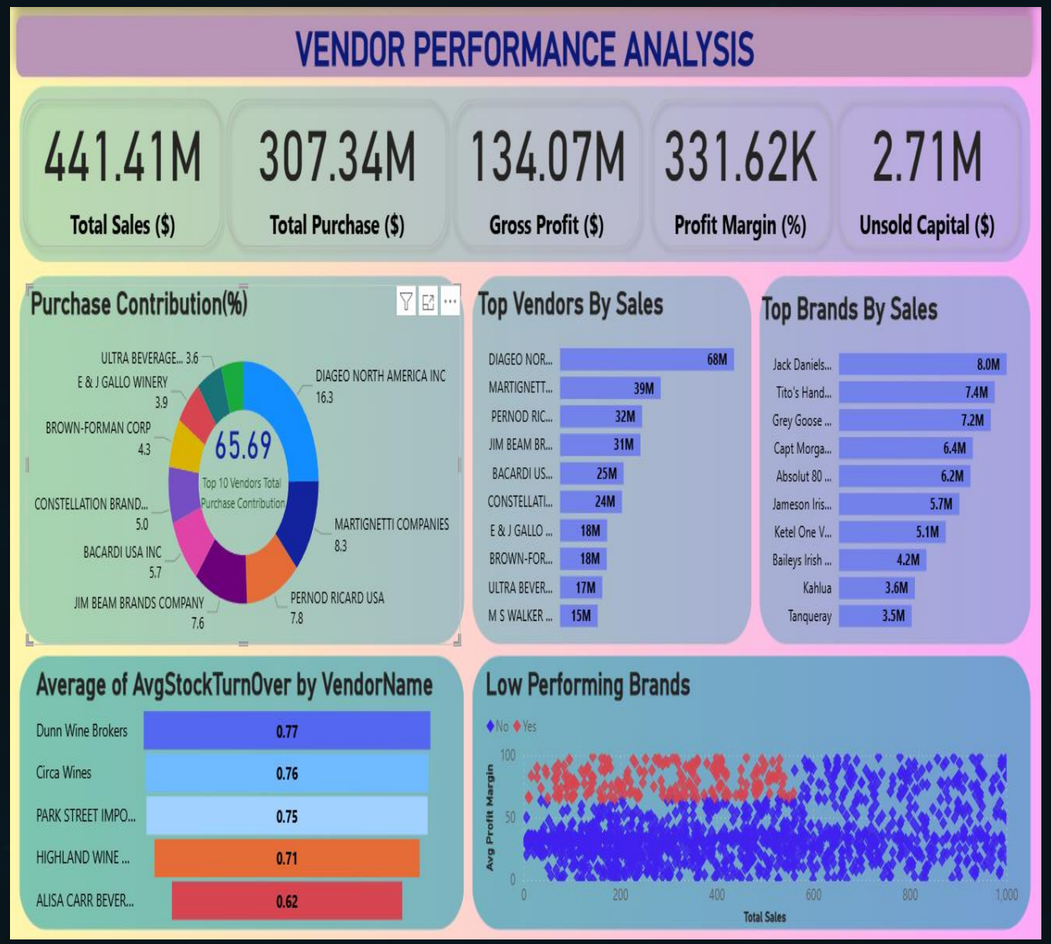
Top 10 vendors contribute 65.69% of purchases (e.g., DIAGEO, Martignetti, Pernod Ricard). High reliance suggests diversification is needed.

Brand Performance

Jack Daniels, Tito's, Grey Goose are top sellers. Low-performing brands with poor margins need product rationalization or discount adjustments.

Inventory & Stock Efficiency

Vendors like Dunn Wine Brokers show high turnover. Alisa Carr Beverage shows lower turnover, indicating potential overstocking.



Strategic Recommendations

1

Diversify Vendors

Reduce dependency on a few large vendors by adding more medium-level partners to mitigate supply risks.

2

Improve Inventory Planning

Increase stock for fast-selling brands and reduce for slow-movers to avoid overstocking and wasted capital.

3

Focus on Best-Selling Brands

Promote top performers to quickly boost total sales and profits.

4

Fix Low-Performing Brands

Implement discounts, offers, or better positioning. Consider reducing or removing if performance doesn't improve.

5

Negotiate Better

Use performance data to negotiate better prices, discounts, or payment terms with top vendors.